

Buyback Debt Coverage and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Buyback Debt Coverage (BDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on BDC achieves an annualized gross (net) Sharpe ratio of 0.38 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 15 (15) bps/month with a t-statistic of 1.90 (2.04), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 16 bps/month with a t-statistic of 2.82.

1 Introduction

Asset prices reflect investors’ marginal utility across states of nature, with risk premia emerging from the covariance between asset payoffs and stochastic discount factors that price consumption streams. While traditional asset pricing models focus on market beta and firm characteristics, mounting evidence suggests that corporate financial policies contain information about firms’ exposure to systematic consumption risk. We identify a significant gap in understanding how firms’ debt-financed share repurchase activities relate to their fundamental risk exposures and expected returns. The ratio of buyback expenditures to outstanding debt—which we term Buyback Debt Coverage (BDC)—potentially captures firms’ implicit assessment of their own consumption risk exposure and their optimal response to time-varying investment opportunities.

Within a consumption-based asset pricing framework, firms with high BDC ratios signal lower exposure to aggregate consumption risk through their revealed preferences in capital allocation decisions. When firms choose to lever up for share repurchases, they implicitly reveal confidence in their ability to service debt obligations across different consumption states, suggesting lower covariance with the marginal utility of consumption (Cochrane et al., 2023). This behavior becomes particularly pronounced during periods of low aggregate consumption volatility, when the price of consumption risk is compressed and firms rationally substitute equity for debt financing (Bansal and Yaron, 2004). The state-dependent nature of buyback activity reflects firms’ optimal response to variation in the intertemporal marginal rate of substitution. During good economic times characterized by low consumption volatility and disaster risk, high BDC firms load negatively on long-run consumption growth, as their aggressive financial policies leave them vulnerable to sudden shifts in consumption dynamics (Wachter, 2013). Moreover, firms engaging in debt-financed buybacks effectively increase their exposure to habit formation risk, as their heightened

leverage amplifies sensitivity to slow-moving consumption trends that drive time-variation in risk premia (Campbell and Cochrane, 1999). The consumption-based interpretation suggests that BDC captures cross-sectional variation in firms' disaster risk exposure, with high BDC firms exhibiting lower sensitivity to rare consumption disasters due to their revealed preference for financial risk over operational hedging.

Our empirical analysis reveals that a value-weighted long-short trading strategy based on BDC achieves an annualized gross Sharpe ratio of 0.38, with the strategy earning average monthly returns of 24 basis points (t -statistic = 2.77). The strategy delivers positive and significant alphas across all standard factor models, with the monthly alpha relative to the Fama-French six-factor model equaling 15 basis points (t -statistic = 1.90). Most notably, when we control for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the BDC strategy maintains a monthly alpha of 16 basis points with a t -statistic of 2.82, demonstrating its unique information content. The economic significance of BDC remains robust across different portfolio construction methodologies and size groups. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the BDC strategy achieves average returns of 22 basis points per month (t -statistic = 2.35), with six-factor alphas ranging from 14 to 29 basis points per month. After accounting for transaction costs using the composite bid-ask spread measure, the net Sharpe ratio equals 0.33, placing the strategy in the top 7% of all documented anomalies. The strategy's factor loadings reveal significant positive exposure to the profitability (RMW) factor with a coefficient of 0.28 (t -statistic = 8.02), consistent with high BDC firms maintaining strong operational performance that supports their aggressive financial policies.

This paper contributes to three distinct strands of the asset pricing literature by establishing BDC as a novel predictor of cross-sectional returns through a consumption-based risk channel. First, we extend the corporate payout literature beyond tradi-

tional measures by demonstrating that debt-financed buybacks contain unique information about consumption risk exposure not captured by existing payout proxies (Grullon and Michaely, 2002; Fama and French, 2001). While prior work focuses on the signaling and agency implications of repurchases, we show that the debt coverage dimension reveals firms’ consumption risk assessment through their financing choices. Second, our findings contribute to the broader literature on financing decisions and expected returns by documenting that BDC subsumes information in related measures such as net equity financing and share issuance (Daniel and Titman, 2006; Pontiff and Woodgate, 2008). The orthogonality of BDC to these established predictors, evidenced by the significant alpha of 16 basis points when controlling for six related anomalies, suggests that debt-financed buybacks capture a distinct aspect of firms’ risk profiles. Third, we advance the consumption-based asset pricing literature by providing empirical evidence that corporate financial policies reflect managers’ implicit assessment of their firms’ exposure to long-run consumption risks and rare disasters (Bansal et al., 2005; Gabaix, 2012). The state-dependent nature of the BDC premium, particularly its strength during periods of low consumption volatility, supports theoretical predictions about the pricing of consumption risk in incomplete markets. Our results have important implications for portfolio management, as the BDC strategy’s net Sharpe ratio of 0.33 and its ability to expand the mean-variance frontier even after controlling for transaction costs demonstrate its economic value beyond academic interest.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several

decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of stock repurchase activity and interest obligations across firms. Buyback Debt Coverage signal relies on two key financial statement items. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the cash flow statement. This variable captures the cash outflow associated with share buyback programs, including both open market repurchases and tender offers. Interest expense (XINT) measures the total interest and related expenses incurred on debt obligations during the fiscal year, as reported in the income statement. This includes interest on short-term and long-term debt, capital leases, and other interest-bearing liabilities. We construct the Buyback Debt Coverage signal by dividing stock repurchases (PRSTKC) by interest expense (XINT) for each firm-year observation. This ratio captures the relationship between a firm’s capital distribution through share repurchases and its debt servicing obligations. Economically, the signal measures how many dollars a firm spends on buybacks for each dollar of interest expense, providing insight into the firm’s capital allocation priorities and financial flexibility. We calculate the signal using fiscal year-end values and require non-missing, positive values for interest expense to ensure meaningful ratios. Firms with zero or missing stock repurchases receive a signal value of zero, while we exclude firm-years with missing or non-positive interest expense from our analysis.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDC signal. Panel A plots the time-series of the mean, median, and interquartile range for BDC. On average, the cross-sectional mean (median) BDC is 16.51 (0.03) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDC data. The signal’s

interquartile range spans -0.00 to 3.87. Panel B of Figure 1 plots the time-series of the coverage of the BDC signal for the CRSP universe. On average, the BDC signal is available for 5.62% of CRSP names, which on average make up 7.02% of total market capitalization.

4 Does BDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDC portfolio and sells the low BDC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDC strategy earns an average return of 0.24% per month with a t-statistic of 2.77. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.14% to 0.31% per month and have t-statistics exceeding 1.81 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 8.02 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 475 stocks and an average market capitalization of at least \$949 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 1.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-21bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.21. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDC trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-five cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the BDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDC, as well as average returns and alphas for long/short trading BDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the BDC strategy achieves an average return of 22 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the BDC strategy relative to the five most common factor models range from 14 to 29 bps/month with t-statistics between 1.48 and 3.05.

5 How does BDC perform relative to the zoo?

Figure 2 puts the performance of BDC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the BDC strategy falls in the distribution. The BDC strategy’s gross (net) Sharpe ratio of 0.38 (0.33) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDC strategy (red line).² Ignoring trading costs, a \$1 invested in the BDC strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$3.36 which ranks the BDC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDC strategy would have yielded \$2.51 which ranks the BDC strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDC relative to those. Panel A shows that the BDC strategy gross alphas fall between the 50 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDC ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of BDC with 209 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDC or at least to weaken the power BDC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDC}BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDC signal in these Fama-MacBeth regressions exceed 1.12,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on BDC is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDC strategy earns alphas that range from 8-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.10, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDC trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.82.

7 Does BDC add relative to the whole zoo?

Finally, we can ask how much adding BDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes BDC grows to \$1489.58.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDC is available.

8 Conclusion

This study provides comprehensive evidence on the asset pricing implications of Buy-back Debt Coverage (BDC) as a predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BDC exhibits meaningful predictive power for future returns, though the economic magnitude and statistical significance are moderate. The value-weighted long/short strategy based on BDC generates an annualized Sharpe ratio of 0.38 gross and 0.33 net of transaction costs, indicating that the signal retains economic value even after accounting for implementation frictions. The strategy delivers monthly abnormal returns of 15 basis points relative to the Fama-French five-factor model augmented with momentum, with marginal statistical significance (t-statistics ranging from 1.90 to 2.04). Notably, BDC demonstrates incremental predictive power beyond existing payout and financing-related factors, generating a statistically significant alpha of 16 basis points per month (t-statistic of 2.82) when controlling for six closely related strategies including share repurchases, net equity financing, and various issuance measures. These findings suggest that BDC captures a distinct dimension of firm financing behavior that is not fully subsumed by existing factors in the literature. From a practical perspective, while BDC offers positive risk-adjusted returns, the moderate Sharpe ratios and alphas suggest it may be most valuable as a complementary signal within a multi-factor framework rather than as a standalone strategy. The signal's ability to generate positive returns net of transaction costs indicates potential implementability for institutional investors, though the economic magnitudes may be insufficient to overcome additional real-world frictions such as market impact and short-sale constraints for large-scale implementations. Several limitations of this study warrant consideration and point to avenues for future research. First, our analysis focuses exclusively on U.S. equity markets, and the generalizability of BDC's predictive power to international markets remains unex-

plored. Second, while we document the statistical properties of BDC, the economic mechanisms driving its return predictability deserve further theoretical and empirical investigation. Understanding whether BDC proxies for financial constraints, managerial signaling, or market timing behavior would enhance our interpretation of the results. Third, the time-varying nature of BDC's predictive power and its performance across different market regimes and business cycles merits additional scrutiny. Future research could examine whether the signal's efficacy varies with monetary policy conditions, credit market dynamics, or aggregate buyback activity. Additionally, investigating the interaction between BDC and firm characteristics such as size, profitability, and growth opportunities could reveal important heterogeneity in the signal's effectiveness. Finally, extending the analysis to incorporate machine learning techniques or non-linear specifications might uncover additional predictive content in the relationship between buyback debt coverage and future returns.

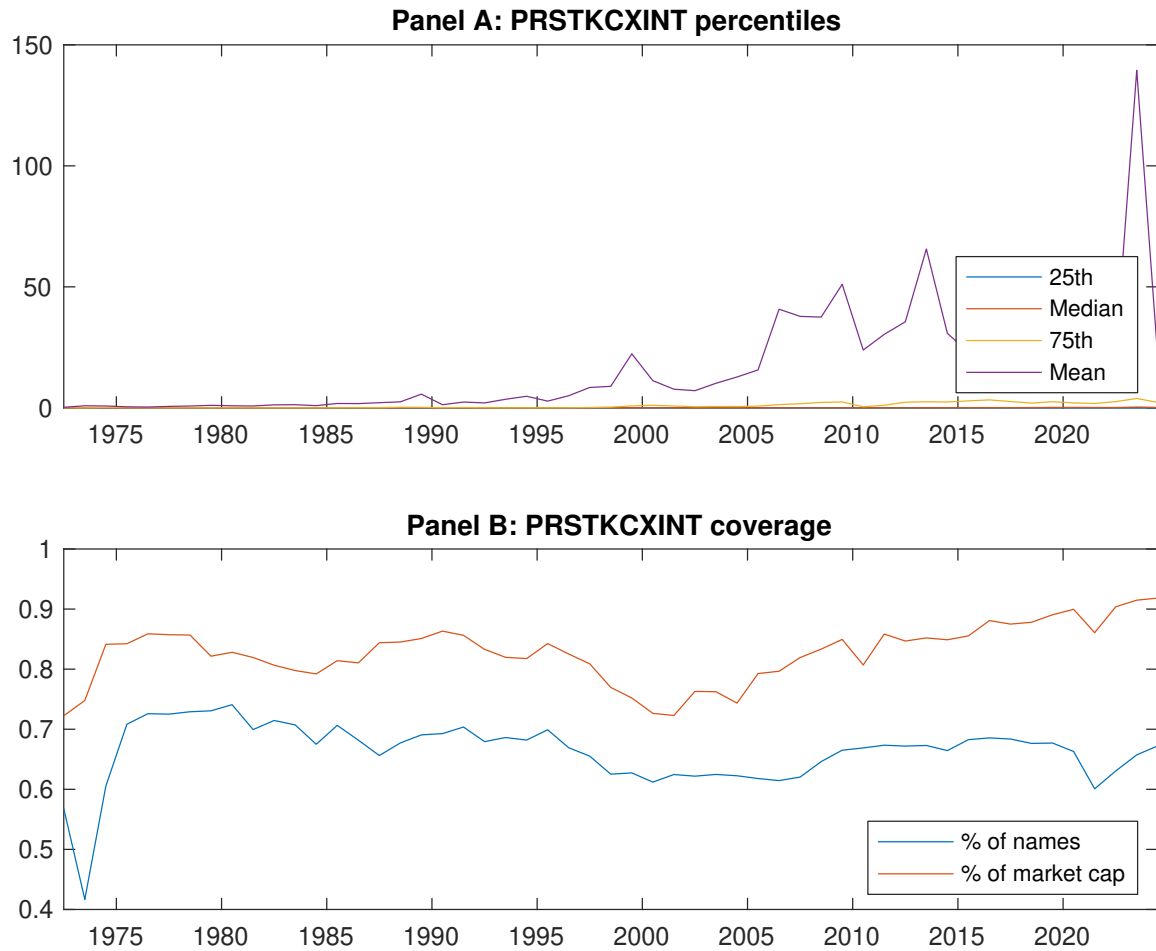


Figure 1: Times series of BDC percentiles and coverage.
This figure plots descriptive statistics for BDC. Panel A shows cross-sectional percentiles of BDC over the sample. Panel B plots the monthly coverage of BDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.66]	0.56 [2.74]	0.52 [2.69]	0.65 [3.77]	0.78 [4.38]	0.24 [2.77]
α_{CAPM}	-0.12 [-1.87]	-0.10 [-1.75]	-0.10 [-1.68]	0.09 [1.74]	0.20 [4.21]	0.31 [3.74]
α_{FF3}	-0.07 [-1.14]	-0.11 [-2.01]	-0.18 [-3.40]	0.03 [0.62]	0.22 [5.04]	0.29 [3.71]
α_{FF4}	-0.06 [-1.04]	-0.12 [-2.18]	-0.17 [-3.16]	0.03 [0.55]	0.22 [4.94]	0.29 [3.58]
α_{FF5}	-0.01 [-0.09]	-0.07 [-1.19]	-0.26 [-4.83]	-0.07 [-1.64]	0.13 [3.11]	0.14 [1.81]
α_{FF6}	-0.01 [-0.10]	-0.08 [-1.40]	-0.25 [-4.53]	-0.07 [-1.48]	0.14 [3.25]	0.15 [1.90]
Panel B: Fama and French (2018) 6-factor model loadings for BDC-sorted portfolios						
β_{MKT}	1.01 [71.77]	1.03 [77.63]	1.05 [82.78]	0.97 [91.51]	0.98 [97.34]	-0.03 [-1.70]
β_{SMB}	0.11 [5.32]	0.15 [7.69]	-0.00 [-0.06]	-0.07 [-4.45]	-0.07 [-4.92]	-0.18 [-6.90]
β_{HML}	-0.09 [-3.41]	0.03 [1.20]	0.17 [7.17]	0.09 [4.26]	-0.14 [-7.17]	-0.05 [-1.34]
β_{RMW}	-0.11 [-3.90]	-0.11 [-4.40]	0.19 [7.71]	0.19 [9.39]	0.17 [8.90]	0.28 [8.02]
β_{CMA}	-0.11 [-2.71]	-0.03 [-0.79]	0.06 [1.50]	0.17 [5.56]	0.15 [5.16]	0.26 [4.99]
β_{UMD}	0.00 [0.12]	0.02 [1.40]	-0.02 [-1.50]	-0.01 [-0.86]	-0.01 [-1.13]	-0.01 [-0.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	822	760	587	475	546	
me (\$10 ⁶)	1356	949	1465	2212	4326	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24 [2.77]	0.31 [3.74]	0.29 [3.71]	0.29 [3.58]	0.14 [1.81]	0.15 [1.90]
Quintile	NYSE	EW	0.32 [3.46]	0.40 [4.50]	0.38 [4.69]	0.31 [3.85]	0.24 [3.03]	0.19 [2.40]
Quintile	Name	VW	0.17 [1.78]	0.26 [2.77]	0.22 [2.48]	0.22 [2.48]	0.04 [0.45]	0.06 [0.66]
Quintile	Cap	VW	0.23 [2.52]	0.28 [3.19]	0.30 [3.62]	0.31 [3.69]	0.19 [2.33]	0.21 [2.52]
Decile	NYSE	VW	0.17 [1.53]	0.23 [2.04]	0.23 [2.17]	0.24 [2.24]	0.11 [1.03]	0.13 [1.21]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.37]	0.30 [3.54]	0.27 [3.44]	0.27 [3.41]	0.15 [1.98]	0.15 [2.04]
Quintile	NYSE	EW	0.12 [1.21]	0.21 [2.17]	0.17 [2.00]	0.14 [1.63]	0.04 [0.43]	0.02 [0.18]
Quintile	Name	VW	0.14 [1.41]	0.25 [2.72]	0.20 [2.36]	0.21 [2.40]	0.06 [0.77]	0.07 [0.90]
Quintile	Cap	VW	0.19 [2.16]	0.26 [2.97]	0.27 [3.25]	0.28 [3.33]	0.19 [2.27]	0.20 [2.39]
Decile	NYSE	VW	0.13 [1.16]	0.22 [1.96]	0.21 [1.99]	0.22 [2.07]	0.12 [1.11]	0.13 [1.21]

Table 3: Conditional sort on size and BDC

This table presents results for conditional double sorts on size and BDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDC and short stocks with low BDC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	BDC Quintiles					BDC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.64 [2.31]	0.60 [2.07]	0.71 [2.48]	0.73 [2.55]	0.82 [3.34]	0.18 [1.99]	0.25 [2.75]	0.20 [2.26]	0.18 [2.01]	0.12 [1.39]	0.11 [1.24]
	(2)	0.64 [2.39]	0.64 [2.38]	0.68 [2.58]	0.85 [3.60]	0.81 [3.58]	0.18 [1.91]	0.30 [3.49]	0.24 [2.92]	0.25 [2.92]	0.10 [1.22]	0.12 [1.41]
	(3)	0.64 [2.60]	0.58 [2.34]	0.75 [3.27]	0.87 [4.00]	0.91 [4.17]	0.28 [2.92]	0.36 [3.87]	0.32 [3.56]	0.34 [3.70]	0.20 [2.31]	0.23 [2.53]
	(4)	0.57 [2.55]	0.66 [2.89]	0.67 [3.19]	0.74 [3.59]	0.93 [4.49]	0.36 [4.26]	0.41 [4.87]	0.39 [4.61]	0.41 [4.74]	0.33 [3.92]	0.35 [4.07]
	(5)	0.51 [2.59]	0.46 [2.42]	0.61 [3.46]	0.73 [4.30]	0.73 [3.93]	0.22 [2.35]	0.26 [2.76]	0.26 [2.84]	0.29 [3.05]	0.14 [1.48]	0.17 [1.81]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	BDC Quintiles					BDC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	352	351	351	353	357	24	24	25	28	32	
	(2)	98	98	97	98	98	50	50	51	52	51	
	(3)	70	71	70	71	71	90	92	93	95	94	
	(4)	61	61	61	61	61	202	209	210	221	218	
(5)	56	56	56	56	56	1368	1278	1459	1699	2594		

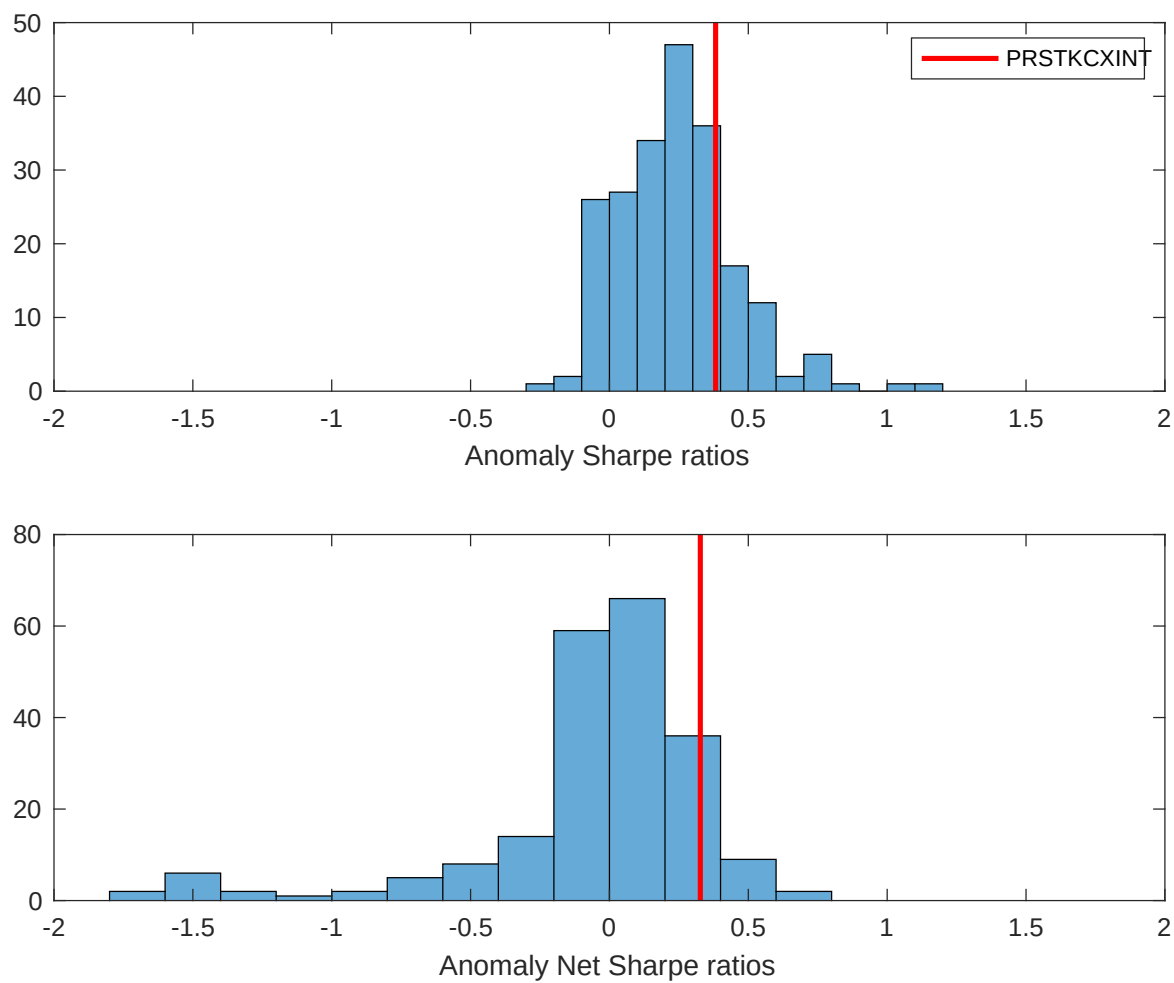


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

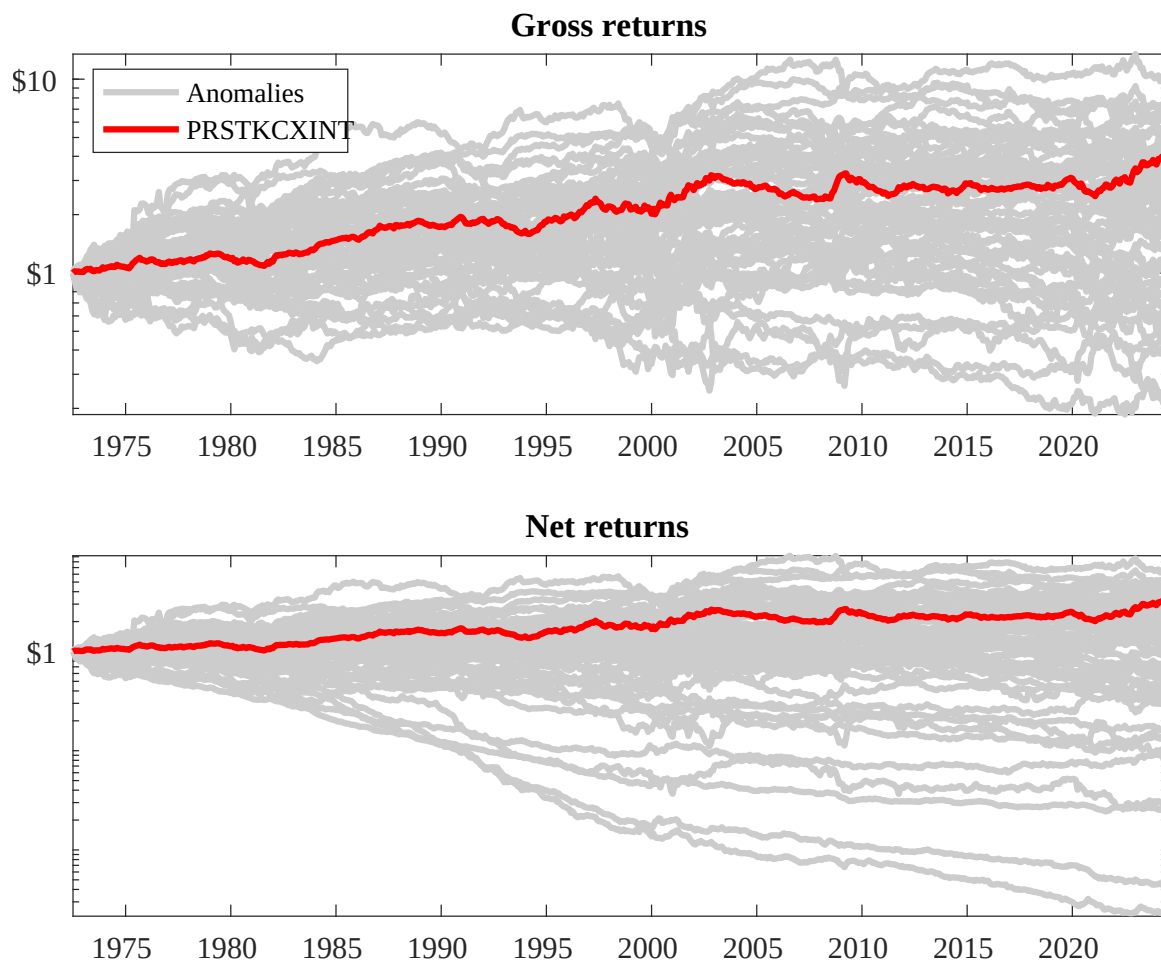
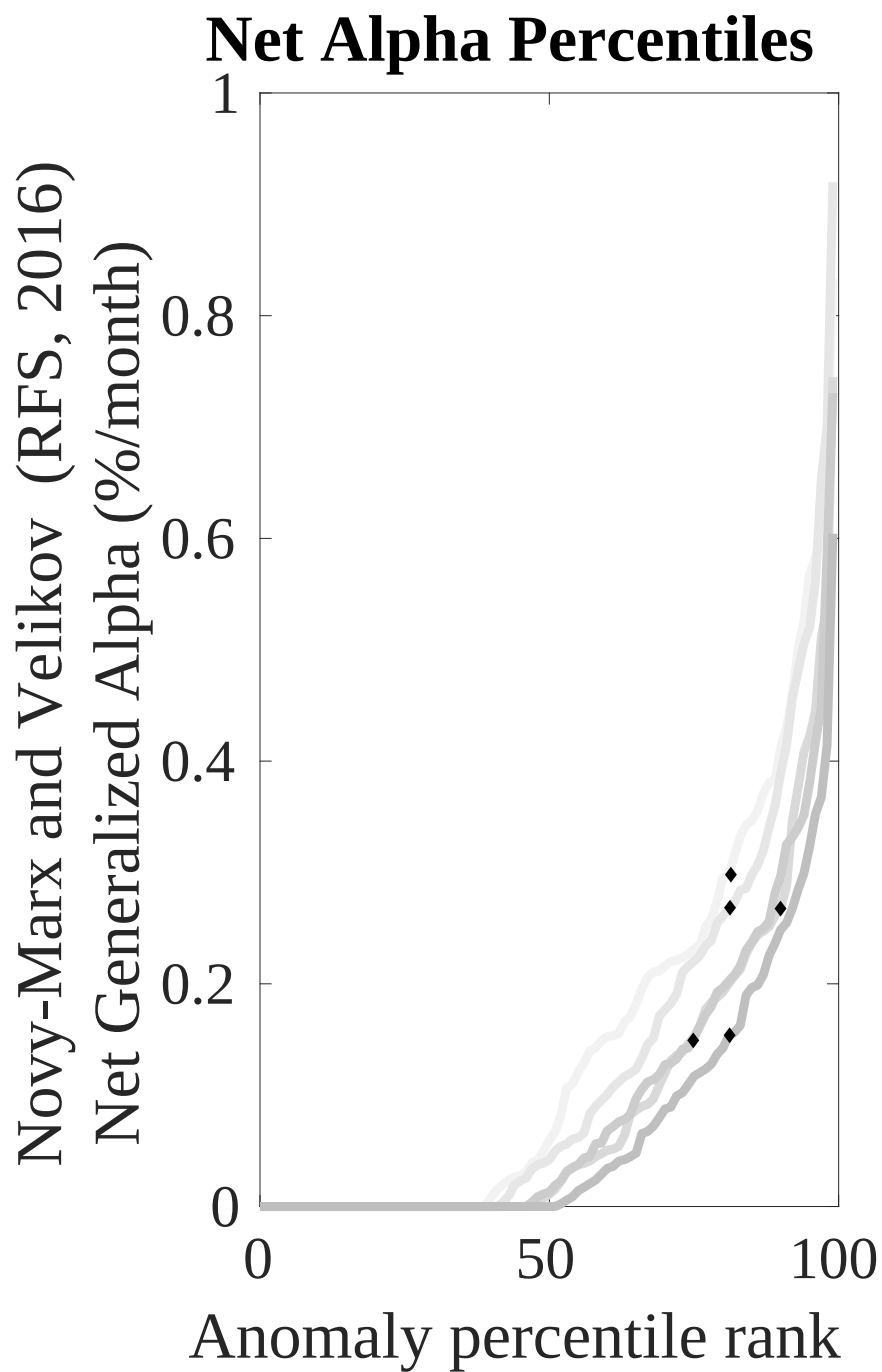
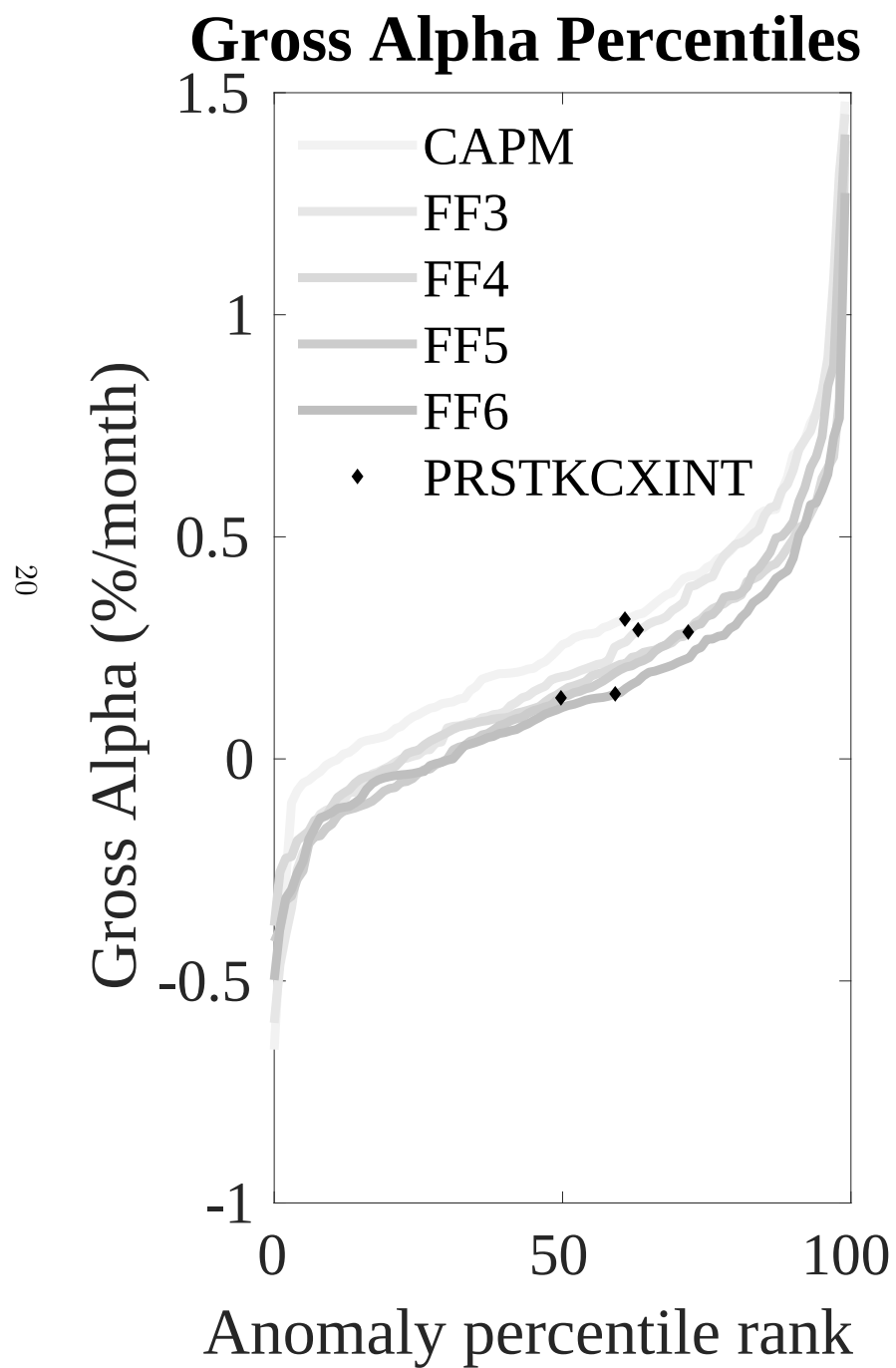


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



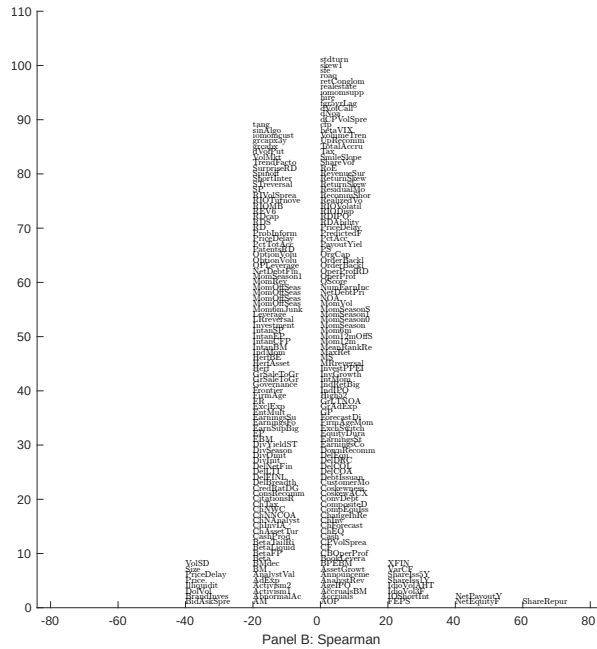
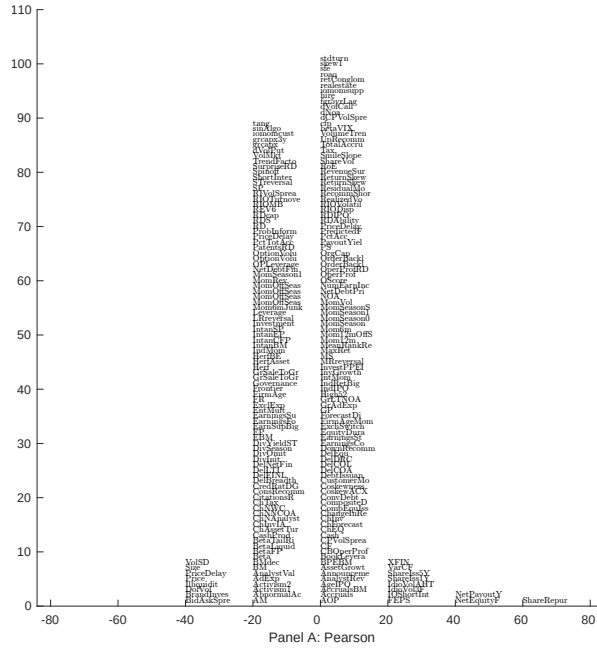


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with BDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

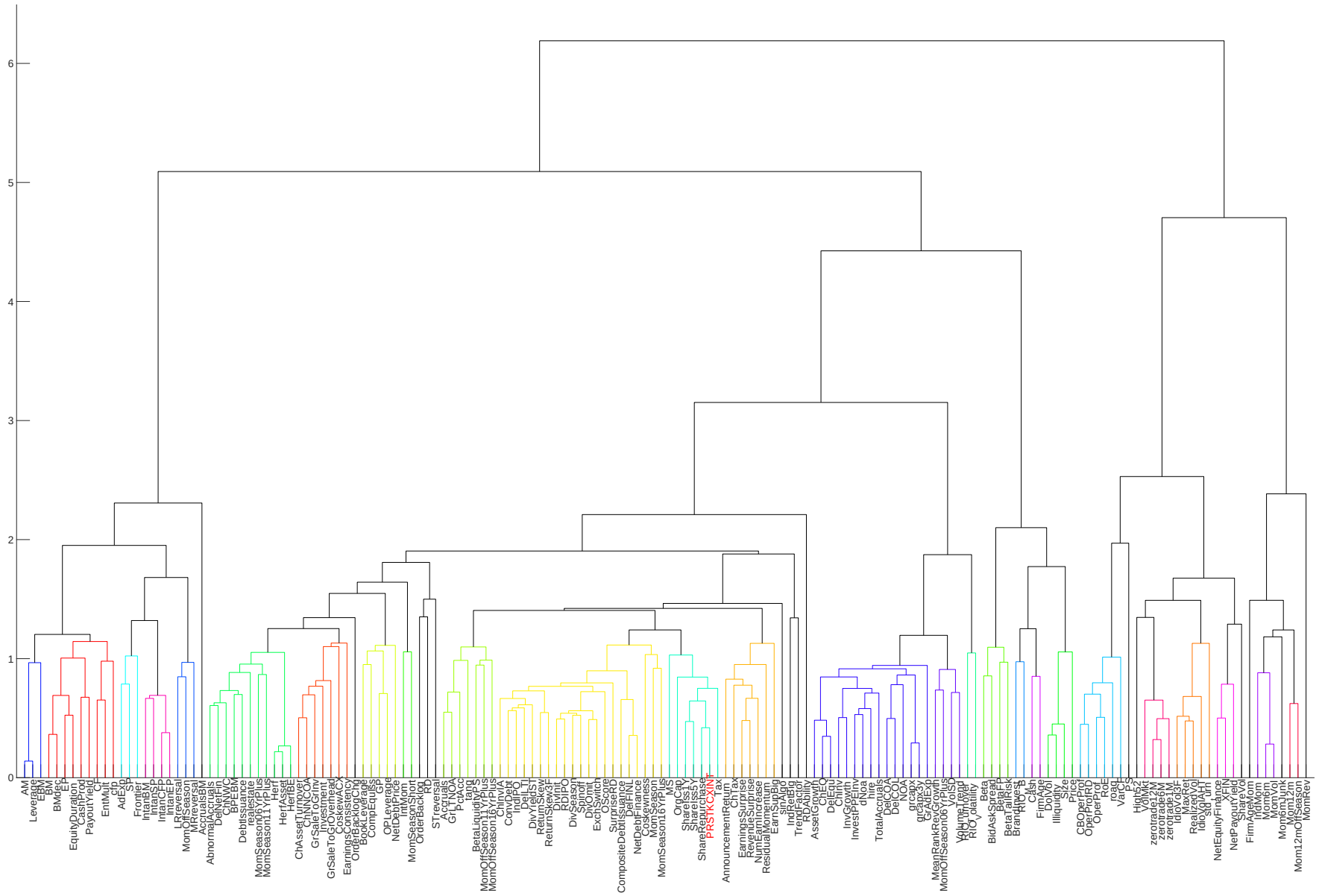


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

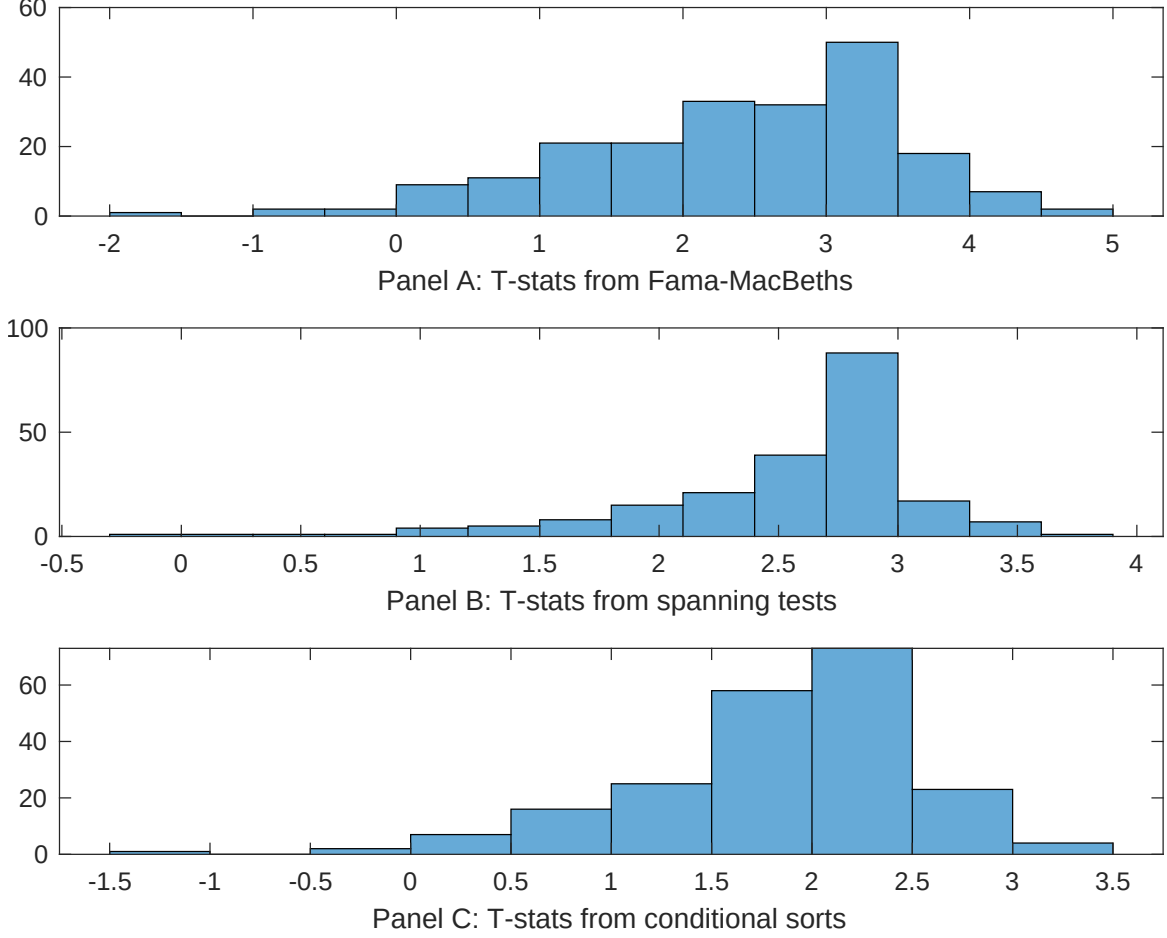


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDC}BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDC}BDC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10 [3.78]	0.13 [5.17]	0.13 [5.34]	0.12 [5.09]	0.13 [5.24]	0.14 [5.94]	0.13 [5.80]
BDC	0.44 [1.12]	0.78 [1.67]	0.89 [2.16]	0.60 [1.39]	0.43 [1.12]	0.12 [2.35]	0.38 [0.67]
Anomaly 1	0.25 [2.79]						0.48 [0.96]
Anomaly 2		0.20 [3.28]					-0.28 [-3.15]
Anomaly 3			0.18 [9.04]				0.55 [2.54]
Anomaly 4				0.29 [5.33]			0.13 [2.40]
Anomaly 5					0.21 [7.70]		0.14 [5.85]
Anomaly 6						0.34 [7.54]	0.28 [6.08]
# months	624	624	619	619	624	619	619
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.18 [2.93]	0.18 [2.65]	0.13 [1.90]	0.17 [2.30]	0.12 [1.69]	0.08 [1.10]	0.16 [2.82]
Anomaly 1	58.99 [18.24]						44.67 [12.91]
Anomaly 2		43.57 [14.50]					17.08 [4.37]
Anomaly 3			38.10 [10.64]				5.47 [1.40]
Anomaly 4				26.15 [9.14]			4.09 [1.44]
Anomaly 5					33.65 [9.25]		10.39 [2.97]
Anomaly 6						28.02 [6.93]	-1.60 [-0.42]
mkt	-2.29 [-1.59]	4.37 [2.68]	-2.45 [-1.50]	1.25 [0.71]	1.67 [0.95]	-2.18 [-1.27]	2.63 [1.79]
smb	-8.61 [-3.88]	-0.82 [-0.32]	-16.15 [-6.57]	-12.75 [-4.97]	-7.20 [-2.60]	-16.20 [-6.28]	0.00 [0.00]
hml	-10.93 [-3.92]	-8.76 [-2.94]	-5.62 [-1.79]	-15.07 [-4.44]	-0.72 [-0.22]	-2.62 [-0.80]	-12.07 [-4.28]
rmw	9.56 [3.20]	1.21 [0.34]	4.77 [1.24]	10.86 [2.91]	7.81 [1.98]	16.06 [4.31]	-8.45 [-2.44]
cma	12.67 [2.98]	-0.54 [-0.11]	4.11 [0.79]	5.76 [1.08]	3.78 [0.70]	14.19 [2.69]	-5.89 [-1.28]
umd	-1.60 [-1.12]	-2.31 [-1.51]	-3.22 [-1.98]	0.07 [0.04]	-1.79 [-1.08]	-3.05 [-1.78]	-1.96 [-1.45]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	53	46	40	37	37	34	60

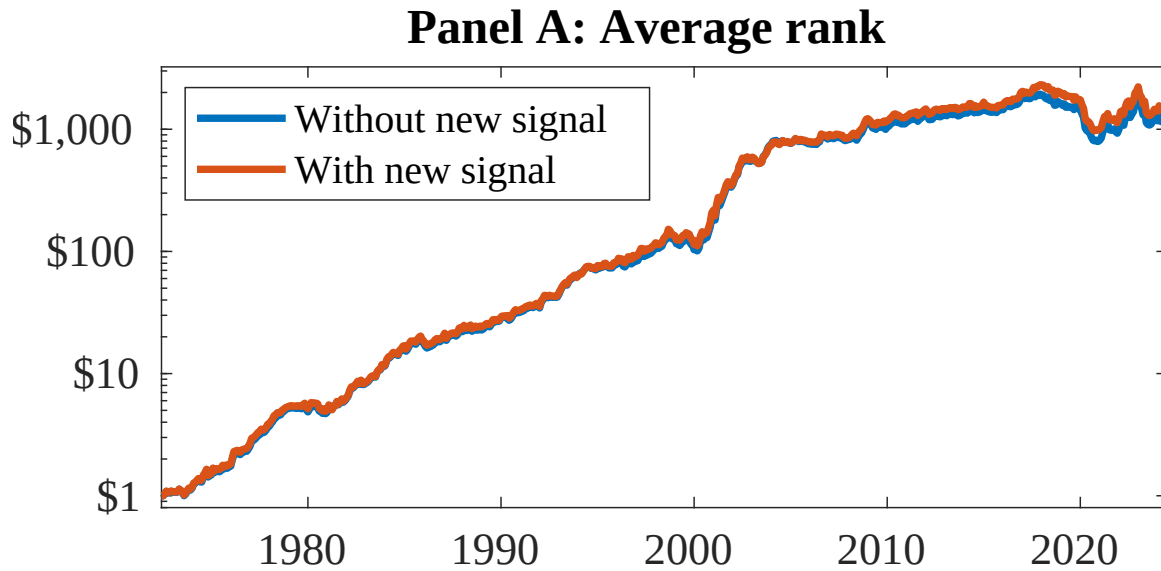


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R., Dittmar, R. F., and Lundblad, C. T. (2005). Consumption, dividends, and the cross section of equity returns. *Journal of Finance*, 60(4):1639–1672.
- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H., Piazzesi, M., and Tuzel, S. (2023). Consumption-based asset pricing with investment shocks. *Journal of Finance*, 78(4):2089–2139.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2001). Disappearing dividends: changing firm characteristics or lower propensity to pay? *Journal of Financial Economics*, 60(1):3–43.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Grullon, G. and Michaely, R. (2002). Dividends, share repurchases, and the substitution hypothesis. *Journal of Finance*, 57(4):1649–1684.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.
- Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.